

CAPSTONE BACKEND DELIVERABLE

SmartStock Database Design

SQLite schema, entity relationships, field definitions, integrity rules, and initialization strategy.

Project	SmartStock Inventory and Sales Management System
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Document	Database Design and Table Specification
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1. Design Overview

SmartStock uses a normalized SQLite database for a small retail inventory and sales workflow. The schema separates authentication, catalog, transactions, operational orders, customers, and workspace configuration. Foreign keys protect the two critical ownership relationships: users to sessions and products to sales.

Database engine	SQLite using Node.js node:sqlite DatabaseSync
Database file	server/database/smartstock.db
Journal mode	WAL
Foreign keys	Enabled with PRAGMA foreign_keys = ON
Initialization	Tables and sample records are created automatically on first start

2. Entity Relationship Diagram

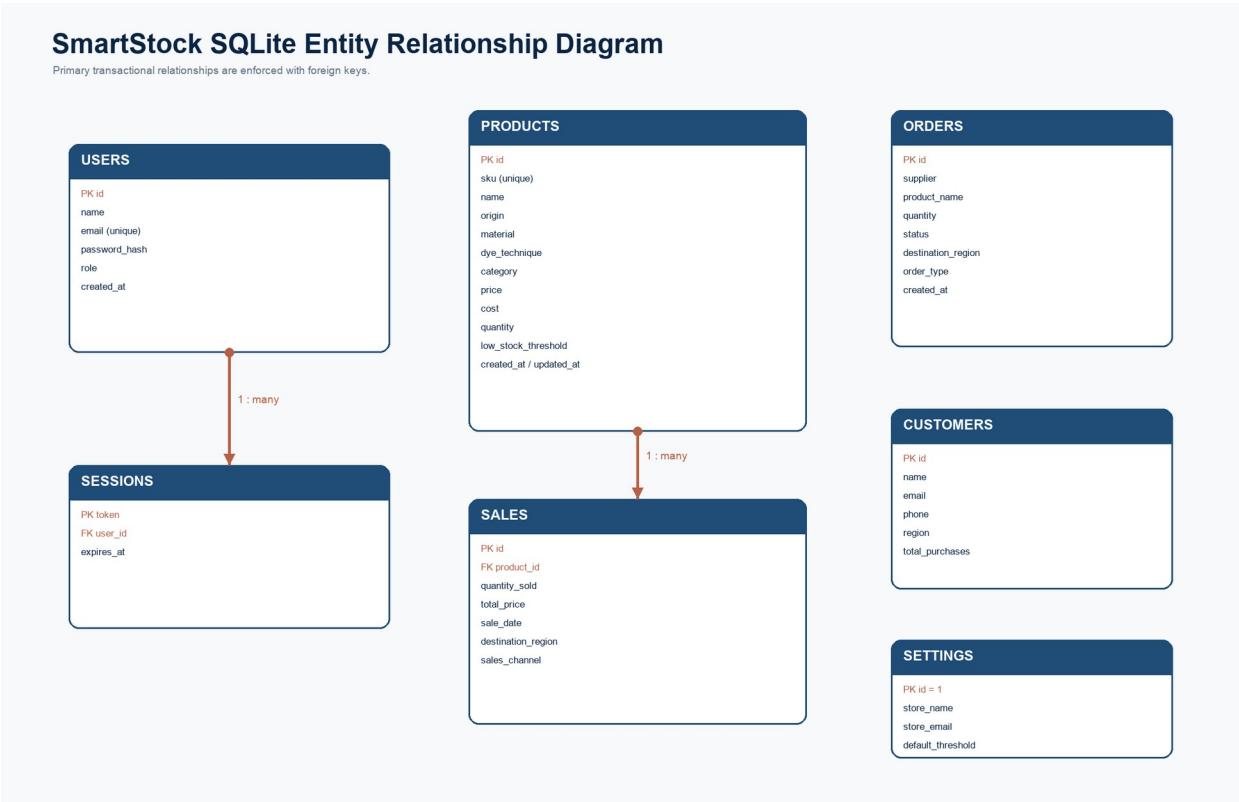


Figure 1. SmartStock database entities and enforced one-to-many relationships.

3. Table Catalog

Table	Purpose	Relationship
users	Authentication identity and role	users.id -> sessions.user_id
sessions	Bearer-token sessions	sessions.user_id -> users.id
products	Catalog and inventory	products.id -> sales.product_id
sales	Sales ledger and regional/channel data	sales.product_id -> products.id
orders	Purchase and international order tracking	Standalone operational record
customers	Customer directory and lifetime value	Standalone directory record
settings	Singleton workspace configuration	Fixed primary key id = 1

4. Detailed Table Definitions

4.1 users

Field	Type	Constraints	Purpose
id	INTEGER	PK, auto increment	Unique user identifier
name	TEXT	NOT NULL	User display name
email	TEXT	NOT NULL, UNIQUE	Login email
password_hash	TEXT	NOT NULL	Salted script password hash
role	TEXT	CHECK admin/employee	Authorization role
created_at	TEXT	NOT NULL	Creation timestamp

4.2 sessions

Field	Type	Constraints	Purpose
token	TEXT	PK	Opaque bearer token
user_id	INTEGER	FK -> users.id	Session owner
expires_at	TEXT	NOT NULL	Session expiration timestamp

4.3 products

Field	Type	Constraints	Purpose
id	INTEGER	PK, auto increment	Product identifier
sku	TEXT	UNIQUE when present	Catalog SKU
name	TEXT	NOT NULL	Product name
origin	TEXT		Craft origin
material	TEXT		Primary material
dye_technique	TEXT		Tie-dye technique

category	TEXT		Product category
price	REAL	NOT NULL, default 0	Retail price
cost	REAL	NOT NULL, default 0	Unit cost
quantity	INTEGER	NOT NULL, default 0	Current stock
low_stock_threshold	INTEGER	NOT NULL, default 5	Alert threshold
created_at	TEXT	NOT NULL	Creation timestamp
updated_at	TEXT	NOT NULL	Last update timestamp

4.4 sales

Field	Type	Constraints	Purpose
id	INTEGER	PK, auto increment	Sale identifier
product_id	INTEGER	FK -> products.id	Sold product
quantity_sold	INTEGER	NOT NULL	Units sold
total_price	REAL	NOT NULL	Server-calculated total
sale_date	TEXT	NOT NULL	Transaction timestamp
destination_region	TEXT	default United States	Customer region
sales_channel	TEXT	default Online Store	Sales channel

4.5 orders

Field	Type	Constraints	Purpose
id	INTEGER	PK, auto increment	Order identifier
supplier	TEXT	NOT NULL	Supplier or sales channel
product_name	TEXT	NOT NULL	Order product
quantity	INTEGER	NOT NULL	Order quantity

status	TEXT	NOT NULL	Operational status
destination_region	TEXT	default United States	Destination
order_type	TEXT	default Purchase	Purchase or International
created_at	TEXT	NOT NULL	Creation timestamp

4.6 customers

Field	Type	Constraints	Purpose
id	INTEGER	PK, auto increment	Customer identifier
name	TEXT	NOT NULL	Customer name
email	TEXT	NOT NULL	Customer email
phone	TEXT		Customer phone
region	TEXT	default United States	Customer region
total_purchases	REAL	NOT NULL, default 0	Lifetime purchase value

4.7 settings

Field	Type	Constraints	Purpose
id	INTEGER	PK, fixed at 1	Singleton settings record
store_name	TEXT	NOT NULL	Store name
store_email	TEXT	NOT NULL	Store email
default_threshold	INTEGER	NOT NULL	Default alert threshold
admin_name	TEXT	NOT NULL	Administrative contact
currency	TEXT	NOT NULL	Display currency

5. Relationships and Integrity Rules

users 1 -> many sessions	A session belongs to one user. Session rows are deleted when their user is deleted.
products 1 -> many sales	Every sale references one product. Product deletion is blocked when sales history exists.
Unique SKU	A partial unique index prevents duplicate non-null SKU values.
Role constraint	User roles are limited to admin or employee.
Transactional sale	Sale insertion and inventory reduction execute inside one database transaction.
Stock validation	Sales are rejected when the requested quantity is not available.
Singleton settings	The settings table uses a CHECK constraint requiring id = 1.

6. Security and Data Handling

- Passwords are never stored as plaintext; each password uses a random salt and script hash.
- Bearer tokens are generated with cryptographically secure random bytes.
- Sessions have an explicit expiration timestamp and can be revoked through logout.
- SQL statements use prepared parameters rather than string concatenation for user data.
- The automated test suite uses a separate temporary database file and does not modify production demonstration data.

7. Initialization and Migration Strategy

On application startup, the database module creates missing tables, enables foreign keys and WAL mode, and applies additive column migrations for fields introduced after the initial schema. Seed users, tie-dye products, orders, customers, and settings are inserted only when their corresponding tables are empty.

```
{
  "admin_demo": "admin@smartstock.com / admin123",
  "employee_demo": "employee@smartstock.com / employee123",
  "automatic_database_path": "server/database/smartstock.db",
  "test_override": "SMARTSTOCK_DB_PATH=/temporary/path.db"
}
```